Demo Institute of Technology
CS201 - Data Structures and Algorithms
Midterm Examination - Synthetic Practice Paper
Academic Year 2024-25 | Time Allowed: 90 minutes | Maximum Marks: 50
SYNTHETIC PAPER: generated for development testing, not an official examination.

SECTION A - Short Answer

1. Define a stack and list two of its primary operations. [5 marks]

2. Explain the difference between an array and a linked list with respect to the cost of inserting an element in the middle. [5 marks]

3. State the time complexity of binary search on a sorted array of n elements and justify your answer. [4 marks]

SECTION B - Problem Solving

4. Given the array [3, 1, 4, 1, 5, 9, 2, 6], show the state of the array after each pass of insertion sort. [8 marks]

5. Derive the worst-case recurrence relation for merge sort and solve it. [10 marks]

6. A binary tree has 15 nodes. What is the maximum possible height of the tree? Justify your answer. [8 marks]

7. Write pseudocode to detect a cycle in a singly linked list. [10 marks]